

Auteur: Siebe Mees
Studierichting: Informatica
Oefeningenreeks 7C nr. 1.5

$$f : \mathbb{R}^3 \rightarrow \mathbb{R} : (x, y, z) \mapsto \cos(xy) + \sin(yz)$$

$$\frac{\partial f}{\partial x} = -y \sin(xy)$$

$$\frac{\partial f}{\partial y} = -x \sin(xy) + z \cos(yz)$$

$$\frac{\partial f}{\partial z} = y \cos(yz)$$

$$\frac{\partial^2 f}{\partial x^2} = -y^2 \cos(xy)$$

$$\frac{\partial^2 f}{\partial x \partial y} = -\sin(xy) - xy \cos(xy)$$

$$\frac{\partial^2 f}{\partial x \partial z} = 0$$

$$\frac{\partial^2 f}{\partial y^2} = -x^2 \cos(xy) - z^2 \sin(yz)$$

$$\frac{\partial^2 f}{\partial y \partial x} = -\sin(xy) - xy \cos(xy)$$

$$\frac{\partial^2 f}{\partial y \partial z} = \cos(yz) - yz \sin(yz)$$

$$\frac{\partial^2 f}{\partial z^2} = -y^2 \sin(yz)$$

$$\frac{\partial^2 f}{\partial z \partial x} = 0$$

$$\frac{\partial^2 f}{\partial z \partial y} = \cos(yz) - yz \sin(yz)$$

References